

A Project Based Seminar Report

on

”Title of Seminar”

Submitted to the

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In partial fulfillment for the award of the Degree of

Bachelor of Engineering

in

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by

Student Name

(Seat No/Roll No& TE)

Under the guidance of

Prof.Guide Name



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Semester-VI, Third Year Engineering

2018-2019



CERTIFICATE

This is to certify that the project based seminar report entitled "**Title of Seminar**" being submitted by **Student Name (Seat No /Roll No and TE IT)** is a record of bonafide work carried out by him/her under the supervision and guidance of **Prof. Guide Name** in partial fulfillment of the requirement for **TE (Information Technology Engineering) - 2015 Course** of Savitribai Phule Pune University, Pune in the academic year 2018-2019.

Date: 31/03/2019

Place: Pune

Prof. Guide Name
Seminar Guide

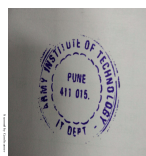
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This Project Based Seminar report has been examined by us as per the Savitribai Phule Pune University, Pune requirements at Army Institute of Technology, Pune-411015 on

Internal Examiner

External Examiner



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(Students Name & Signature)

Abstract

The word abstract comes from the Latin abstractum, which means a condensed form of a longer piece of writing. There are two main types of abstract: the (1) Descriptive and the (2) Informative abstract. The type of abstract you write depends on your discipline area.

A good abstract:

1. uses one well-developed paragraph that is coherent and concise, and is able to stand alone as a unit of information.
2. covers all the essential academic elements of the full-length paper, namely the background, purpose, focus, methods, results and conclusions.
3. contains no information not included in the paper.
4. is written in plain English and is understandable to a wider audience, as well as to your discipline-specific audience.
5. often uses passive structures in order to report on findings, focusing on the issues rather than people.
6. uses the language of the original paper, often in a more simplified form for the more general reader.
7. usually does not include any referencing.
8. in publications such as journals, it is found at the beginning of the text, while in academic assignments, it is placed on a separate preliminary page

You may follow the steps given below for writing an abstract:

1. First re-read your seminar report for an overview. Then read each section and condense the information in each down to 1-2 sentences.
2. Next read these sentences again to ensure that they cover the major points in your seminar report.
3. Ensure you have written something for each of the key points outlined above for either the descriptive or informative abstract.

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Chapter 1

INTRODUCTION TO PROJECT TOPIC

1.1 INTRODUCTION TO PROJECT TOPIC

1.1.1 Introduction to Project

Students are expected to write brief introduction of the project topic. This section may be common for all the students of that group. However students can have different approach in explaining their project.

Since we are not going to get separate project report from the students, it is good to have one or two pages common for all the students of that project. Once this project introduction and aim objectives of explained, students can start with actual seminar content writing.

1.1.2 Motivation behind project topic

Students are expected to write " Why they thought about this project. They could also explain the " Explain need of the project". Any statement which motivated to take up this project

1.1.3 Aim and Objective(s) of the work

Project aims are a statement of the overall " Why" of the project. A project aim is a statement starting with the words: " The aim of this project is. . ."

That statement tells the reader what your overall goal is, what it is you want to achieve. It does not go into details or describe specific tasks.

Project objectives tell the "How" of the project aim. You want to achieve your goal and the objectives state how this will be done - what major tasks will be undertaken and what your major targets are. Most projects will have between six and nine objectives. A project objective is usually a statement starting with the words "To..."

Objectives are subsidiary to aims and are the steps you are going to take to answer your problem statement or a specific list of tasks needed to accomplish the goals of the project. This must be highly focused and feasible and should address the more immediate project outcomes. Guide must personnel check these aim and objectives and make students write these statements properly.

1.1.4 Introduction to Seminar Topic

Internal guides need to clearly identify separate topics for individual students while preparing seminars. It should be based on project topic/area.

Chapter 2

Introduction to Index Compression

2.1 LITERATURE SURVEY OF INDEX COMPRESSION

In information retrieval(IR) central data structures are Dictionary & Inverted Index. Compression techniques applied on this central data structures for efficient Information retrieval.

2.1.1 Goal of Compression

- Main goal of compression is conserve disk space & increase speed.
- For faster cache utilization and for faster memory transfer compression and decompression speed must be high.

2.1.2 Benefits of compression

- First we need less disk space due to compression. As the disk space is a valuable resource it is very important to store data in less memory space. usually compression ratio 1:4 is easy to achieve. Potentially cut the cost of storing index upto 75 % .
- Second benefit of index compression is increased use of caching. Search systems use some parts of dictionary and index more. With compression lot of information is stored into main memory. It also decreases the response time of the IR system substantially.
- Third benefit is faster transfer of data from disk to memory. Efficient decompression algorithms work fast so that time of transferring a compressed data is less than uncompressed data.

2.2 Statistical Properties of Terms in IR

- Heaps Law: It is the Estimation of the number of the terms.

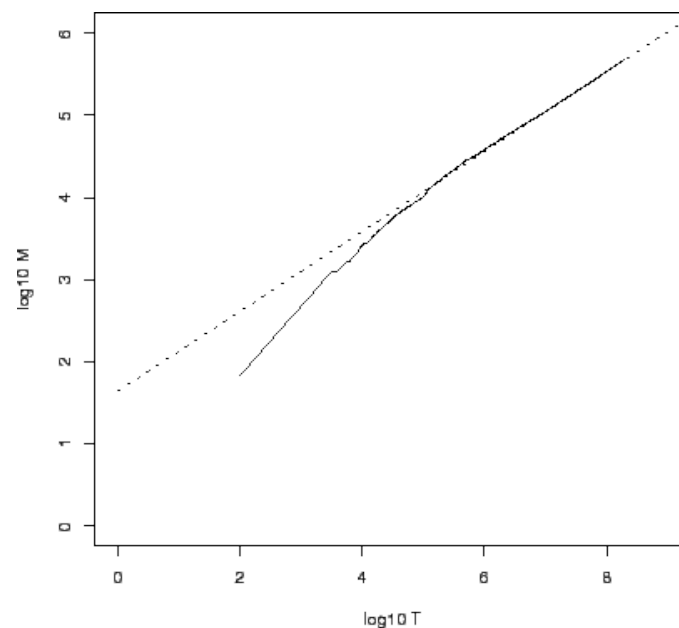


Figure 2.1: Heaps law

- Zipfs Law: It is used for modeling the distribution of terms(occurrences).

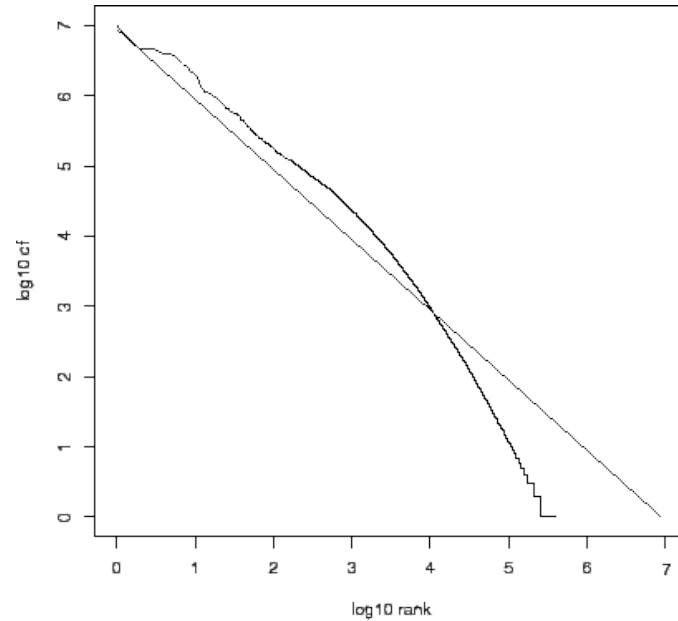


Figure 2.2: Zipfs Law

2.2.1 Lossless Compression

In lossless compression information is preserved. Mainly data stored in this type is easily recovered. Lossless compression is generally used for "discrete" data. Lossless compression algorithms are needed in cases that require compression where we want the reconstruction to be identical to the original.

2.2.2 Lossy Compression

In this type of compression technique it actually discards some information. Lossy compression is used where "lost" information is unlikely ever to be used by the search system. Types are Case Folding, Stemming, Stop word elimination. Eg. Web Search is characterized by a large number of documents. Short queries discard hits far down the list.

2.3 Dictionary Compression

One of the primary factors in determining the response time of an IR system is the number of disk seeks necessary to process a query. Main goal of the dictionary compression is to fit the documents in memory size so that memory seeks should be less in number. There are few types of dictionary compression as follows.

2.3.1 Storing the dictionary as an array of fixed-width entries

Table 2.1: Fixed size storage

Term	Documnet Freq	Pointer
a	656265	—
aachen	65	—
....	..	—
zulu	221	—
20 Bytes	4 Bytes	4 Bytes

Average length of English language is 8 Bytes. For each entry we are storing extra 12 bytes which is wastage of memory. Also there is no way storing words which are more than 20 Bytes long.

2.3.2 Dictionary as a string

We overcome on the disadvantage of fixed size storage by storing the strings as one string of characters. The pointer to next string is stored separate from previous one. This saves 60 % of storage as we are saving 12 Bytes per word in average.

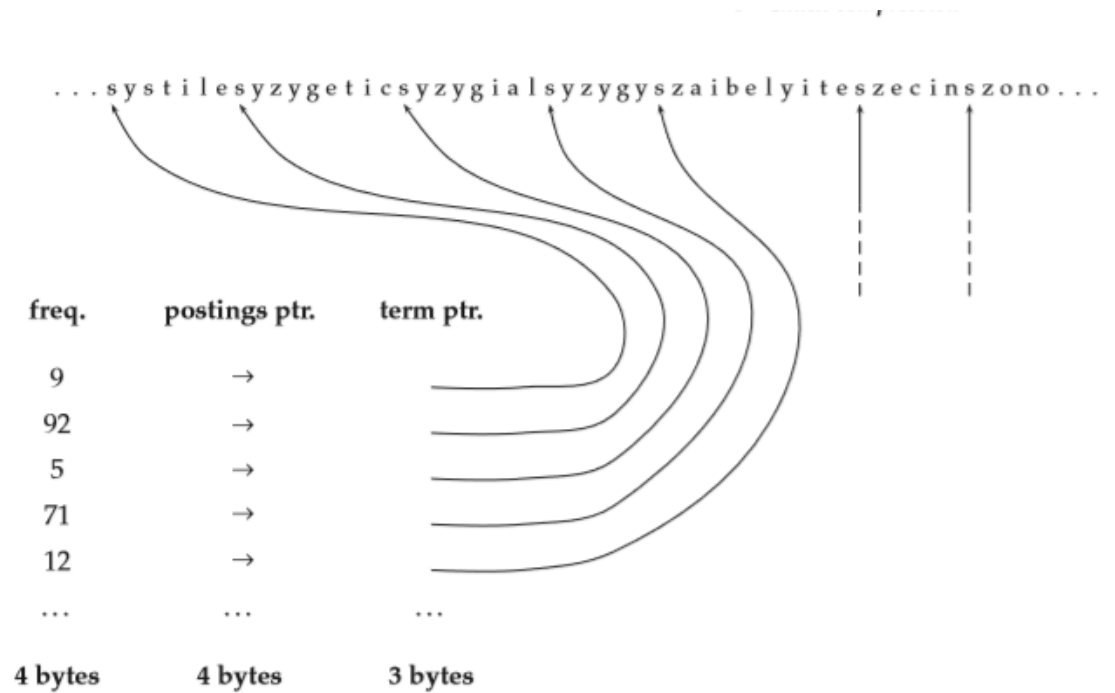


Figure 2.3: Dictionary as String

2.3.3 Blocked Storage

For better compression we group the dictionary by grouping in the terms in the string into blocks of size k and keeping a term pointer for first term of each block. By increasing the block size to k we get better compression. Binary search algorithm is applied on this type of storage and usually takes 25 % of time than uncompressed data.

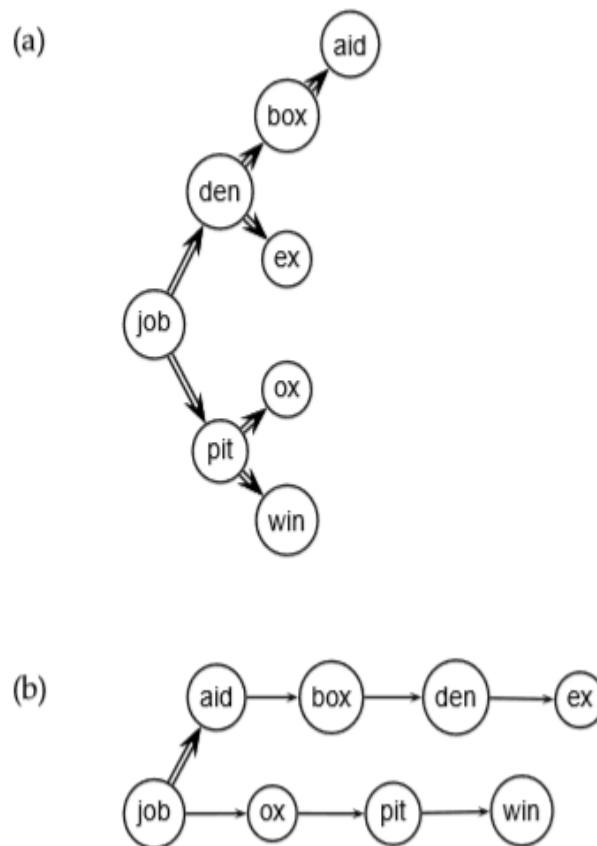


Figure 2.4: Blocked Storage

2.3.4 Front Encoding

In front Encoding a sequence of terms with identical prefix("automat") is encoded by marking the prefix with * and replacing it with o in subsequent terms.

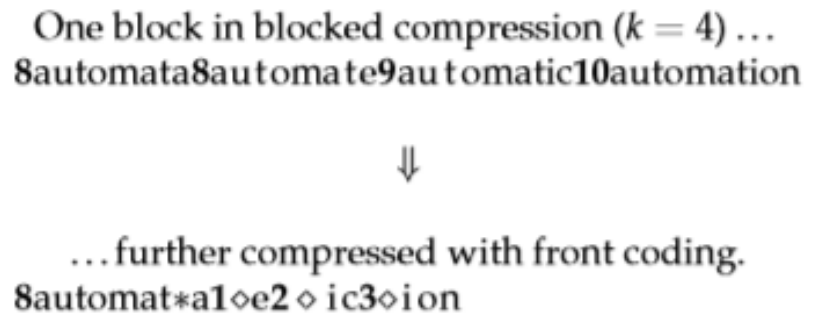


Figure 2.5: front Encoding

2.3.5 Summary

Comparison between all the techniques previously discussed is given below by taking standard comparisons.

Table 2.2: Summary

Data Structure	size in MB
Fixed Size Storage	11.2
Term pointer to String	7.6
Blocked Storage	7.1
Blocked Storage with front coding	5.9

2.4 Posting file compression

Chapter 3

CHAPTER 3

3.1 SEMINAR RELATED OTHER CHAPTERS

All other details of seminar topic chapter-wise if necessary including methodology/algorithms, advantages and disadvantages, applications, enhancements could be added.

Table 3.1: Parameters and their values

Sr. No.	Parameter	Value
1	Simulation Area	5 km X 5 km
2	No. of roaming users	500
3	No. of PBS queries	In a range from 15 to 500
4	User mobility technique	Plus mobility (roaming)
5	Time interval	2 hours
6	Anonymity threshold value	Low (5), Medium (7), High (10)

Chapter 4

n

Chapter 5

CONCLUSION

Conclusions usually serve two functions. The first is to summarize and bring together the main areas covered in the writing, which might be called "looking back" . The second is to give a final comment of your seminar.

For example your could say this seminar is undertaken to explain..... and evaluate This study has found that generally Or In this seminar we... .

Chapter 6

REFERENCES

List all the material used from various sources for making this project proposals

Bibliography

[1] Paper 1

[2] Paper 2

[3] Paper 3

[4] Paper 4